

Kanishka Sivarajah

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EDUCATION

Boston University, College of Engineering

B.S., Electrical Engineering

Boston, MA

Expected May 2027

- **BU University Scholar:** Merit full-need scholarship awarded to those with outstanding academic achievement.

PROJECTS

Quantum Engineering & Photonics Research Project | *CAD Design, Mechanical Support Structures*

- Designed custom 3D models using OnShape for structural supports and protective housing for experimental quantum photonics components.
- Collaborated with the research team to create manufacturable designs compatible with the continuous-wave diamond maser experiment.

LED Catcher Game | *Verilog, Vivado, FPGA, Finite State Machine*

- Developed standalone reaction-based game on FPGA board, where players score by activating corresponding switch as a moving LED reaches its position across a 16-LED array.
- Designed finite state machine to control game logic, implementing edge detection, custom miss detection, score tracking, and automatic reset after three missed catches.
- Built multiplexed 7-segment display system to provide real-time score and 3 s pause after successful catches.
- Organized workload and created PowerPoint presentation to present details of project.

Truss Analysis | *MATLAB, Structural Analysis, Experimental Testing*

- Designed acrylic truss structure to achieve maximum load-to-weight efficiency (>32 oz capacity).
- Developed MATLAB model to predict maximum load capacity and identify first member to buckle.
- Oversaw project execution from design through testing, organizing deliverables, tracking progress, and supporting data-driven design decisions.

Gait Tracker | *Arduino, Temperature Sensors, 3D Modeling, 3D Printing, Circuit Assembly*

- Developed device using proximity sensors to measure gait, using red LED for toe strikes, blue LED for heel strikes, and haptic buzzer for simultaneous strikes
- Created custom 3D models for the device housing; 3D printed components; soldered circuit components.
- Directed a multidisciplinary team of 4 engineers, facilitated team meetings, communicated deliverable deadlines, created presentations, and worked on technical report.

SKILLS

Hardware/Design: Arduino, Soldering, 3D Printing, FPGA, Laser Cutting, Drill Press, OnShape, KiCAD

Programming: MATLAB, Python, HTML, CSS, JavaScript, Cura, Arduino IDE, Vivado, Linux, Microsoft Office

EXPERIENCE

Boston University Initiative for Literacy Development (BUILD)

Boston, MA

Teaching Assistant

Oct 2025 – May 2026

- Assist teachers with in-class activities for class of 32 students.
- Tutor students in math and reading through 1:1 and group sessions, 2x per week.
- Maintain records of student assessments and tutoring activities and attend weekly debrief meetings.

BU College of Engineering Career Development Office

Boston, MA

Office Assistant

Jan 2024 – May 2025

- Managed and organized digital databases of employer job postings and student career resources, ensuring timely and accurate updates.
- Coordinated student and employer communications while maintaining confidential records and supporting office operations.
- Assisted with career fairs and recruiting events by organizing registration data and event logistics.